

UNIVERSITY OF MILANO-BICOCCA

Master's Degree in Data Science

Dense Semantic Segmentation of ECG Signals

A 1D Spatiotemporal U-Net Approach

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Agenda

1. Problem Definition
2. Data Analysis & Engineering
3. Proposed Solution
4. Experimental Results
5. Alternative Experiments & Edge Cases

Problem Definition & Task Analysis

Initial Assessment:
While preliminary guidelines suggested a standard classification paradigm, an in-depth inspection of the PhysioNet dataset [Goldberger et al., 2000] revealed that labels were high-resolution temporal annotations mapping the exact onset and offset of cardiac waves.

Refined Task: Dense Semantic Segmentation

Objective: Predict the underlying physiological cardiac phase at each individual time step (250Hz sampling rate) via a sequence-to-sequence mapping.

Target Classes:

- **Class 0:** N/A (Isoelectric Line / Background)
- **Class 1:** P Wave (Atrial Depolarization)
- **Class 2:** QRS Complex (Ventricular Depolarization)
- **Class 3:** T Wave (Ventricular Repolarization)

Exploratory Data Analysis

1. Extreme Class Imbalance

- The dataset exhibits a profound class imbalance. The background isoelectric line (Class 0) heavily dominates the signal, accounting for over 50% of all time samples.
- Critical morphological features, such as the P-Wave, represent less than 11% of the data. This imbalance poses a significant risk of network collapse into the majority class.

2. Task Complexity and Signal Noise

- Accurately anchoring spatial boundaries across continuous noise presents a highly complex sequence modeling task.
- The network must learn to delineate exact temporal boundaries across signals affected by baseline wander, high-frequency noise, and severe amplitude variations.

Data Engineering Pipeline: MATLAB Pre-processing

The Bottleneck

- Directly importing `QTData.mat` into Python yielded unstructured memory objects due to nested proprietary MATLAB hierarchies.
- Patient signals exhibited varying lengths, complicating batch processing operations.

The MATLAB Data Pipeline

- We engineered an intermediate script using *cell arrays* to dynamically manage variable signal lengths across the 105 patients.
- Target labels were mapped from categorical text tables into numerical `uint8` matrices.
- Exported a sanitized dataset enforcing the `-v7` flag for flawless Python compatibility.

Data Engineering Pipeline: Python Refactoring

Python Environment Setup

- Continuous signals were partitioned into standardized receptive fields of 1000 samples (4 seconds).
- Target arrays were One-Hot Encoded to support Categorical Cross-Entropy loss computation.

Crucial Methodology: Patient-Aware Data Splitting

In medical deep learning, performing a randomized split causes severe *data leakage*. To rigorously prevent this and guarantee unbiased generalization, the dataset partitioning was executed **subject-wise**. Sliding windows were generated strictly within individual patient boundaries, ensuring **absolute zero overlap** of subjects between the $18,858$ Training sets and $4,715$ Testing sets.

Proposed Architecture: 1D Spatiotemporal U-Net

Architectural Objective:
To extract complex morphological features while strictly preserving exact temporal localization, we mapped the sequence problem to a **1D Convolutional U-Net** [Ronneberger et al., 2015].

Topological Key Components:

- **Encoder (Contracting Path):** Blocks of `Conv1D` and `MaxPooling1D` capture hierarchical semantic representations of the cardiac waves.
- **Decoder (Expanding Path):** `UpSampling1D` layers fully restore the original spatial dimensionality.
- **Skip Connections:** Direct concatenations link early spatial features with deep semantic embeddings. *This mitigates the degradation of temporal boundaries induced by pooling operations.*
- **Output Layer:** Dense classification via `softmax` activation, yielding a temporal probability distribution.

Training Strategy & Optimization

1. Class Imbalance Mitigation (Loss Function)

- To counteract the background dominance, we injected dynamic `sample_weights` directly into the Categorical Cross-Entropy loss.
- Weights were calculated inversely proportional to class frequencies, heavily penalizing misclassifications on minority features (e.g., P-Wave weight ≈ 2.35 vs Background ≈ 0.49).

2. Advanced Callback Synergy (Keras API)

- **ReduceLROnPlateau:** Dynamically decayed the Learning Rate upon loss stagnation, facilitating micro-optimizations inside complex local minima.
- **EarlyStopping:** Prevented overfitting by monitoring `val_loss` with a patience of 10 epochs.
- **ModelCheckpoint:** Automatically serialized the optimal weights during training.

Quantitative Evaluation: Test Set Performance

The model was rigorously evaluated on the unseen Patient-Aware Test Set. To reflect the standards of dense segmentation tasks, we integrated spatial overlap metrics alongside standard sample-based scores.

Cardiac Phase	**Precision**	**Recall**	**IoU**	**Dice Coeff.**
0: N/A (Background)	0.96	0.91	0.88	0.94
1: P Wave	0.87	0.95	0.83	0.91
2: QRS Complex	0.95	0.98	0.94	0.97
3: T Wave	0.90	0.94	0.85	0.92
Overall Accuracy		**0.93**		

Quantitative Evaluation: Result Analysis

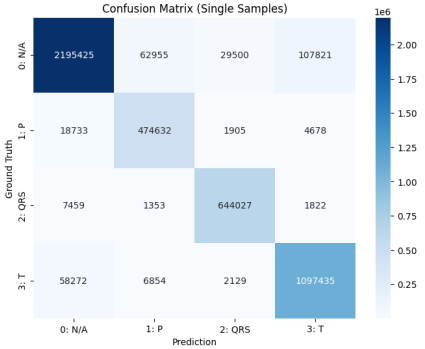
1. Class Collapse Prevented

The weighting strategy successfully secured a $> 94\%$ Recall on critical waves. This demonstrates that the model accurately identifies and isolates minority classes despite the massive background dominance.

2. High Spatial Precision

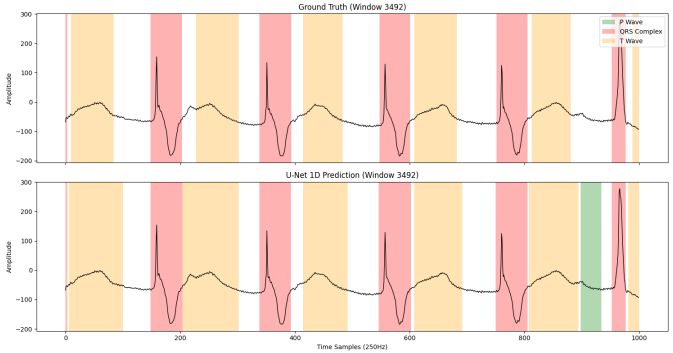
The strong Intersection over Union (IoU) scores confirm the network's outstanding capability to accurately capture the exact morphological boundaries of the cardiac events, going beyond simple presence detection.

Quantitative Evaluation: Confusion Matrix



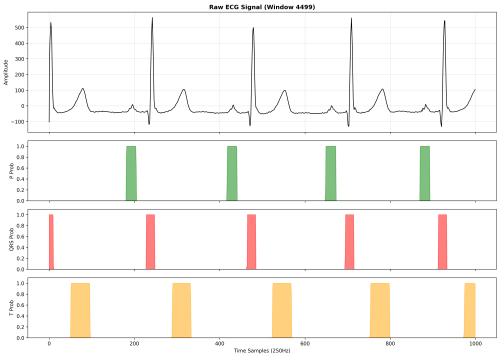
Clinical Interpretation: The matrix highlights robust class discrimination. The primary source of misclassification occurs at the boundary between the T-Wave (3) and the Background (0). This specific error is *physiologically coherent* due to the ambiguous, asymptotic decay of ventricular repolarization.

Qualitative Evaluation: Physiological Alignment



Observation: The predicted segments exhibit millimetric alignment with the Ground Truth. The deep convergence achieved by the U-Net successfully eliminated the high-frequency classification noise (jittering) typically seen in early training epochs, producing clean and continuous semantic regions.

Model Uncertainty Analysis (Softmax Probabilities)



Confidence Analysis: Analyzing the raw Softmax outputs reveals extreme confidence (probabilities near 1.0, sharp rectangular shapes) for the QRS complex. Conversely, the gradual slope of the T-Wave probability (orange) physically mirrors the physiological decay, visually explaining the structural boundary uncertainty.

Ablation Studies & Alternative Configurations

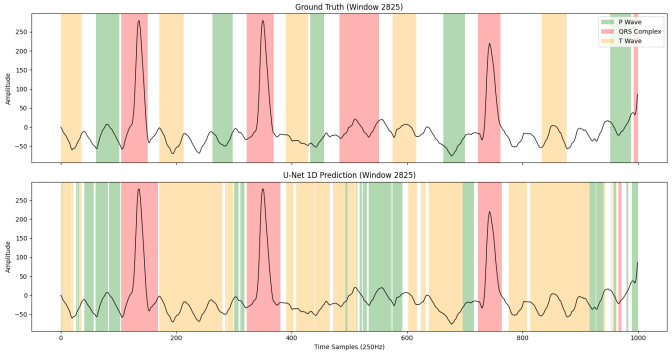
1. Failed Architectural Configuration

- *Standard Conv1D:* Removing the U-Net skip connections caused severe *temporal shifting*.
- Boundary predictions for the QRS complexes were systematically displaced from the physiological peaks due to spatial data loss during pooling operations.

2. Alternative Loss Functions

- *Dice Loss vs CCE:* While theoretically sound for dense segmentation, Dice Loss induced severe gradient instability during early epochs due to the extreme sparsity of the P-Wave.
- Weighted Categorical Cross-Entropy (CCE) proved mathematically more stable and achieved superior convergence.

Hard Example Mining (Edge-Case Analysis)



Failure Analysis: By isolating the absolute worst-performing segment in the Test Set (**Accuracy: 42.10%**), we identify the architecture's limitations. In presence of severe signal artifacts or extreme baseline wander, structural anchoring is heavily corrupted, leading to fragmented segmentation.